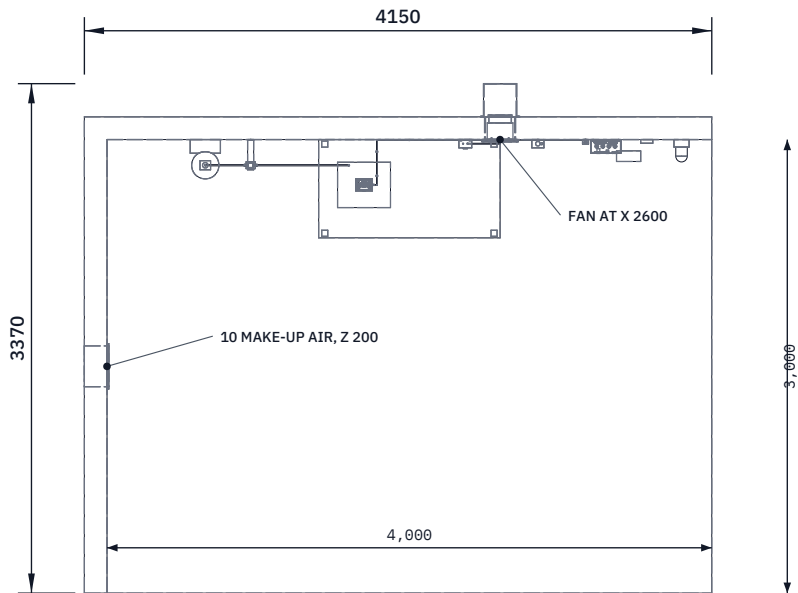


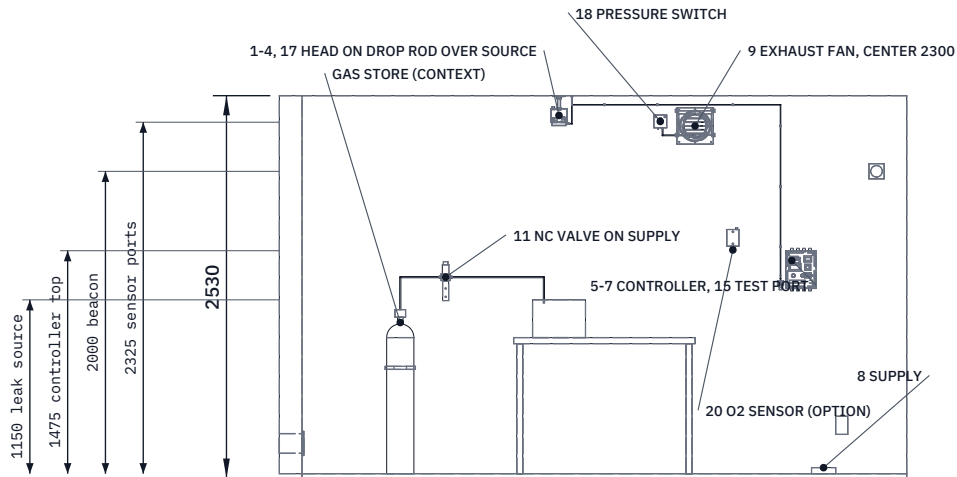
CONCEPT, NOT FOR FABRICATION

H2Guard · github.com/BoujeeEnjinia1701/h2guard



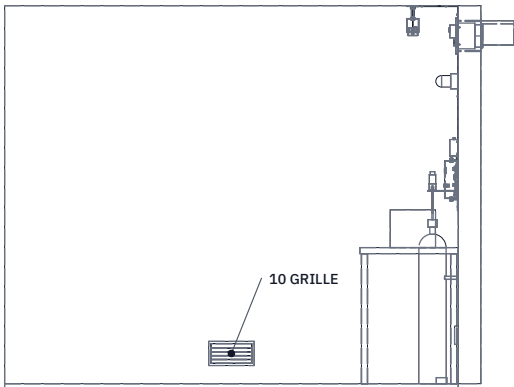
TOP VIEW

Scale 1:50; looking down (along -Z)



FRONT VIEW

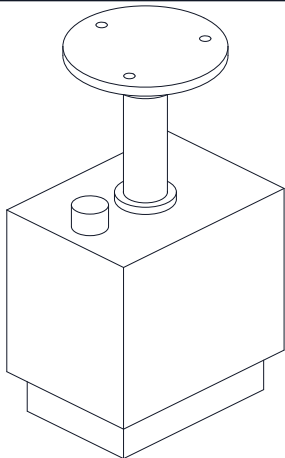
Scale 1:50; looking at the front (along +Y)



RIGHT VIEW

Scale 1:50; looking at the right side (along -X)

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA for TRL 3 (from cad/src/model.py)	2026-09-25	AC
P2	Notes: head placement rule, timed escalation (DDR-002)	2026-09-25	AC
P3	Layout and labels tidied	2026-09-25	AC
P4	Design for construction (DDR-003): brackets, plates, sleeve	2026-10-01	AC
P5	Head on ceiling drop rod; pressure switch; O2 option (DEC-001)	2026-10-02	AC



DETECTOR HEAD ON ITS DROP ROD, ITEMS 1 TO 4 AND 17

Not to scale; ports and bump test cup face down

MAIN DIMENSIONS AND INTERFACES (MM)

Room 4000 x 3000 x 2500 (30 m3); leak source 1150 above floor
Head 110 x 80 x 90 on drop rod; ports 175 below ceiling, 1,175 above source
Arrestor discs 25 x 2; sensor 2.2 behind disc
Bump cup 96 x 66 x 22 (114 mL); tube 4 OD, 3,108 run
Controller 200 x 250 x 90; top 1,025 below ceiling
Fan 150 spigots, body in a 200 wall sleeve; grille top 80 below ceiling
Make-up grille 300 x 160 on far wall, center 200
Valve 1/4 in NC, hydrogen rated, on a wall bracket; gas fitting by others
Ports 0 from apparatus centre (plume radius 141)
Pressure switch at fan; O2 cell 1,492 above floor (option)
Warning held 5 min closes valve and latches (firmware, DDR-002)
Third-angle; front view from -Y (room side); HGD-CAL-001

H2GUARD

github.com/BoujeeEnjinia1701/h2guard

General arrangement in the 30 m3 reference room

DRAWING NO. HGD-DWG-001		REV P5	SHEET 1 of 1	SIZE ANSI B
SCALE 1:50	UNITS mm	PROJECTION 	DATE 2026-10-02	DRAWN Amish Chadha
MATERIAL / NOTES Bought-in parts per bom/bom.csv; room, bench and gas store are context. PRELIMINARY, NOT FOR FABRICATION				
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